

Notes: Filippou, Maurer, Pezzo & Taylor (JFE, 2024)

Importance of Transaction Costs for Asset Allocation in Foreign Exchange Markets

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Currency universe and quotes	2
2.2	Monthly currency returns and trading positions	2
2.3	Proportional transaction costs	3
2.4	Price impact of trading	3
2.5	Volume extrapolation	3
2.6	Strategies	3
3	Models and empirical specifications	4
3.1	Before-cost and after-cost returns	4
3.2	Mean-variance portfolio	4
3.3	Mean-variance-transaction-cost optimization	4
3.4	Fund-size experiments	4
4	Identification and empirical design	5
4.1	Measurement validation	5
4.2	Portfolio-design comparison	5
4.3	What the paper cannot fully prove	5
5	Tables and figures	5
5.1	Table 1: Average trading costs by currency	5
5.2	Figure 1 and Table 2: Proportional costs and realized Amihud validation	6
5.3	Figure 2 and Figure 3: Volume extrapolation and price impact	7
5.4	Table 3: Strategy performance before and after transaction costs	8
5.5	Figure 4 and Figure 5: Cumulative returns and strategy capacity	8
6	Short summary	9
7	Conclusion	10
7.1	Core conclusion	10
7.2	Economic intuition	10
7.3	Takeaway	10

1 Big picture

- **Question.** How important are transaction costs for currency portfolio performance, especially when funds are large enough for their trades to move FX quotes?
- **Core distinction.** The paper separates small proportional costs from bid–ask spreads and large trading-volume price-impact costs caused by executing sizable orders.
- **Main finding.** Proportional costs are modest, but price-impact costs have first-order effects. Many popular FX strategies become unprofitable once assets under management are large enough.
- **Why this is new.** Most FX asset-pricing and currency-strategy papers deduct bid–ask spreads but ignore the price impact created by trading volume. This paper shows that this omission is economically large.
- **Measurement contribution.** The authors build a realized-Amihud-based price impact measure for 26 currencies from 1986 to 2024. They expand scarce CLS volume data by extrapolating trading volume using bid–ask spreads.
- **Strategy set.** The paper evaluates mean–variance portfolios, carry/value/momentum-style strategies, dollar strategies, and characteristic-sorted long–short strategies.
- **Optimization contribution.** The authors implement a mean–variance-transaction-cost portfolio, denoted *MVTC*, that explicitly internalizes trading costs in the optimization.
- **Main performance result.** Without price impact, standard strategies look profitable. With price impact, many are severely impaired; *MVTC* remains robust even for large funds.
- **Capacity result.** Most strategies have low capacity: their after-cost Sharpe ratios fall quickly as AUM grows. *MVTC* has substantially higher capacity because it trades less and chooses cost-aware positions.
- **Economic takeaway.** In FX markets, liquidity is deep but not free. For institutional scale, portfolio design must treat price impact as part of the investment problem rather than as an after-the-fact cost adjustment.

2 Data and measurement

2.1 Currency universe and quotes

- The sample covers 26 currencies against the USD from February 1986 to January 2024.
- The authors use Barclays Bank International and Reuters quotes from Datastream for spot and one-month forward bid, ask, and mid prices.
- They use Olsen high-frequency spot quotes to improve measurement of spot bid–ask spreads.
- CLS order-flow data are available only for 12 currencies and only for September 2012 to September 2021.
- The paper therefore constructs an extrapolation to extend the price-impact panel to more currencies and a longer history.

Interpretation: The paper’s key measurement problem is that good price data are long and broad, but volume data are short and narrow. The econometric contribution is to use reliable price/spread information to infer missing volume.

2.2 Monthly currency returns and trading positions

The one-month realized currency return from an uncovered long position in currency i is

$$r_{i,t+1} = \ln \left(\frac{X_{i,t+1}}{F_{i,t}} \right) = fd_{i,t} + \Delta x_{i,t+1}, \quad (1)$$

where $X_{i,t}$ is the spot exchange rate, $F_{i,t}$ is the one-month forward exchange rate, $fd_{i,t}$ is the forward discount, and $\Delta x_{i,t+1}$ is the spot exchange rate change.

Interpretation: The strategies rebalance monthly, so trading costs are tied to changes in portfolio weights from one month to the next.

2.3 Proportional transaction costs

For each currency and transaction type $\kappa \in \{P^+, P^-, S^-, S^+\}$, the proportional cost is

$$PC_t = \sum_{i=1}^{N_t} \sum_{\kappa \in K} C_{i,t}^{\kappa} \Delta_{i,t}^{\kappa}, \quad (2)$$

where $C_{i,t}^{\kappa}$ is the cost per dollar traded and $\Delta_{i,t}^{\kappa}$ is the fraction of wealth traded.

Interpretation: Bid-ask spread costs are small for many currencies, but they are only one part of total trading cost.

2.4 Price impact of trading

The price-impact cost is

$$PI_t = \sum_{i=1}^{N_t} \sum_{\kappa \in K} \Pi_{i,t}^{\kappa}(\Delta_{i,t}^{\kappa}) \Delta_{i,t}^{\kappa}, \quad (3)$$

where $\Pi_{i,t}^{\kappa}(\cdot)$ maps trade size into quote movement.

The realized Amihud measure for spot transactions is

$$A_{i,t}^S = \frac{RPV_{i,t}}{v_{i,t}^S}, \quad (4)$$

where $RPV_{i,t}$ is realized price variation and $v_{i,t}^S$ is daily USD spot volume.

Interpretation: The Amihud measure converts price variation per dollar traded into a price-impact parameter. Large funds face high costs because order size scales with AUM.

2.5 Volume extrapolation

The cross-sectional volume extrapolation is based on

$$\ln(\bar{v}_i^S) = a + b \ln(\overline{PBA}_i^S) + \epsilon_i. \quad (5)$$

The fitted relationship is strongly negative: currencies with higher average bid-ask spreads have lower trading volume.

Interpretation: This relation lets the authors expand the realized-Amihud panel beyond the 12 currencies and short window for which CLS volume is available.

2.6 Strategies

The paper studies:

- *MV*: mean-variance optimized currency portfolio;
- *MVTC*: mean-variance-transaction-cost optimized portfolio;

- *HML*: high-minus-low carry-style portfolio;
- *RB*: rank-based portfolio;
- *DOL* and *DDOL*: dollar strategies;
- *VAL*: value strategy based on real exchange rates or PPP;
- *MOM*: momentum strategy based on past currency performance.

Interpretation: The strategy set spans optimized portfolios and popular rules of thumb. This allows the paper to show both the problem and a cost-aware solution.

3 Models and empirical specifications

3.1 Before-cost and after-cost returns

Let r'_t be the realized excess return before trading costs. The after-cost realized excess return is

$$r'^{c}_t = r'_t - (1 + r_{US,t-1} + r'_t)(PC_{t-1} + PI_{t-1}). \quad (6)$$

Interpretation: Costs are deducted from wealth and compound with realized returns. This makes transaction costs especially damaging for high-turnover strategies.

3.2 Mean–variance portfolio

The mean–variance portfolio solves

$$\max_{\theta_t} \left\{ \theta'_t \mu_t - \frac{\lambda}{2} \theta'_t V_t \theta_t \right\}, \quad (7)$$

where μ_t is expected currency return and V_t is the covariance matrix.

Interpretation: *MV* targets the optimal before-cost risk-return trade-off. It can require large rebalancing trades and therefore become expensive for large funds.

3.3 Mean–variance-transaction-cost optimization

The *MVTC* portfolio modifies the objective to account for trading costs:

$$\max_{\theta_t} \left\{ \theta'_t \mu_t - \frac{\lambda}{2} \theta'_t V_t \theta_t - TC(\theta_t - \theta_{t-1}; W_t) \right\}. \quad (8)$$

Interpretation: *MVTC* trades off expected return, risk, and the direct cost of moving from the current portfolio to the new portfolio. It sacrifices little before-cost performance but sharply reduces price impact.

3.4 Fund-size experiments

The paper evaluates strategy performance for:

- no price impact;
- initial AUM of USD 100 million;
- initial AUM of USD 1 billion;
- capacities up to USD 5 billion.

Interpretation: The key comparative statics are with respect to AUM. Proportional costs do not depend much on fund size; price impact does.

4 Identification and empirical design

4.1 Measurement validation

- The paper validates its realized-Amihud construction against the Ranaldo and Santucci de Magistris measure.
- Table 2 reports high correlations across alternative measures for the available currencies.
- Figure 2 shows that average volume is strongly related to average bid–ask spreads, supporting cross-sectional extrapolation.

Interpretation: The credibility of the cost estimates rests on showing that the extrapolated liquidity measures behave like observed realized-Amihud measures.

4.2 Portfolio-design comparison

- The authors compare $MVTC$ with the same currency universe and return inputs used by standard strategies.
- The benchmark strategies ignore price impact in portfolio construction.
- $MVTC$ internalizes costs before trading.

Interpretation: The comparison isolates whether transaction-cost-aware design improves performance relative to common before-cost portfolio rules.

4.3 What the paper cannot fully prove

- Volume data must be extrapolated for many currencies and dates.
- The price-impact function is linear, which may understate nonlinear effects for very large trades.
- The analysis abstracts from strategic execution, order-splitting, and dealer-client heterogeneity.
- Internet Appendix robustness checks address many variants, but transaction-cost measurement is inherently difficult.

Interpretation: The paper’s estimates are designed to be conservative. If anything, nonlinear price impact could make the costs even larger for large funds.

5 Tables and figures

5.1 Table 1: Average trading costs by currency

Read:

- The table reports average adjusted bid–ask spreads and average realized Amihud measures for 26 currencies.
- Major currencies such as EUR, JPY, GBP, and CHF have lower proportional costs and deeper markets.
- Emerging-market currencies such as ZAR, HUF, MXN, and BRL tend to have higher spreads and/or price impact.

- The B&R spread data are often much wider than the adjusted Olsen-based spreads, motivating the adjustment.

Economic message: Liquidity differs substantially across currencies. These differences matter because high-cost currencies can dominate trading costs even when their expected returns look attractive before costs.

Table 1
Average trading costs by currency.

Currency i	All days			Days with B&R data coverage				
	N Obs	$\overline{PBA}_i^S$	$\bar{A}_i^S$	N Obs	$\overline{PBA}_i^{S,BR}$	$\overline{PBA}_i^S$	$\bar{A}_i^S$	$\Delta_{iW}^{\text{b}} W$
AUD	9653	4.6	0.019	9638	8.1	4.6	0.019	91.8
BRL	7718	6.9	0.101	7714	7.3	6.9	0.101	1.6
CAD	9653	3.1	0.009	9631	5.4	3.1	0.009	127.8
CHF	9653	3.5	0.038	9653	5.6	3.5	0.038	28.1
CSK	7604	10.4	0.484	7604	11.6	10.4	0.484	1.2
DEM	3107	4.3	0.018	3065	6.4	4.3	0.018	60.1
ESP	3048	7.1	0.032	1261	5.9	5.6	0.035	5.3
EUR	6542	1.5	0.004	6521	3.7	1.5	0.004	293.7
FRF	3107	4.6	0.021	3065	9.6	4.7	0.021	115.6
GBP	9653	2.8	0.011	9652	4.3	2.8	0.011	70.1
GRD	3489	15.1	0.103	1761	9.4	7.7	0.12	7.1
HUF	7849	10.9	0.59	7849	11.4	10.9	0.59	0.4
IEP	3107	10.5	0.064	3065	18.6	10.4	0.064	63.6
ITL	3107	7.4	0.043	3065	11.6	7.4	0.043	47.7
JPY	9653	3.1	0.007	9650	5.9	3.1	0.007	187.8
KRW	8746	5	0.074	7819	6.7	5.2	0.082	8.9
MXN	9021	6.6	0.042	7849	7.8	7	0.048	9.1
NLG	3107	4.7	0.02	3065	6.4	4.7	0.02	41.2
NOK	9653	4.8	0.188	9652	5.8	4.8	0.187	2.9
NZD	9652	7.8	0.068	9604	12.2	7.9	0.068	32.5
PLN	7587	9.4	0.267	7587	10.7	9.4	0.267	2.4
PTE	2246	10.9	0.065	1261	7.9	7.5	0.067	3.2
SEK	9653	5.8	0.156	9652	8.2	5.8	0.156	7.4
SGD	9653	4.4	0.027	9651	7.2	4.4	0.027	51.1
TWD	8837	7.2	0.033	7825	7.8	6.8	0.037	13.8
ZAR	9628	12.7	0.093	8978	25.6	11.5	0.089	78.9

Notes: The table reports trading costs for each currency against the USD. The first three columns use data from our full sample, the last five columns consider the sub sample of days for which B&R data is available. N obs reports the number of daily observations. $\overline{PBA}_i^{S,BR} = \frac{1}{T} \sum_t PBA_{i,t}^{S,BR}$ is the average proportional bid-ask spreads (in bps) computed from B&R data. $\overline{PBA}_i^S = \frac{1}{T} \sum_t PBA_{i,t}^S$ is the average adjusted spread (in bps) according to Eq. (1). $\bar{A}_i^S = \frac{1}{T} \sum_t A_{i,t}^S$ is the average realized Amihud measure

5.2 Figure 1 and Table 2: Proportional costs and realized Amihud validation

Read:

- Figure 1 shows proportional costs for spot and forward markets over 1986–2024.
- Proportional costs generally decline over time but spike during crises, especially in forward markets.
- Table 2 shows high correlations between the authors’ hourly-quote measure and the benchmark Rinaldo–Santucci de Magistris measure.
- The “constant volume” version is also correlated with the benchmark, but the hourly-quote version is generally stronger.

Economic message: Bid–ask costs are measurable and comparatively small, but the validation step is important because the price–impact analysis builds on the same realized–Amihud logic.

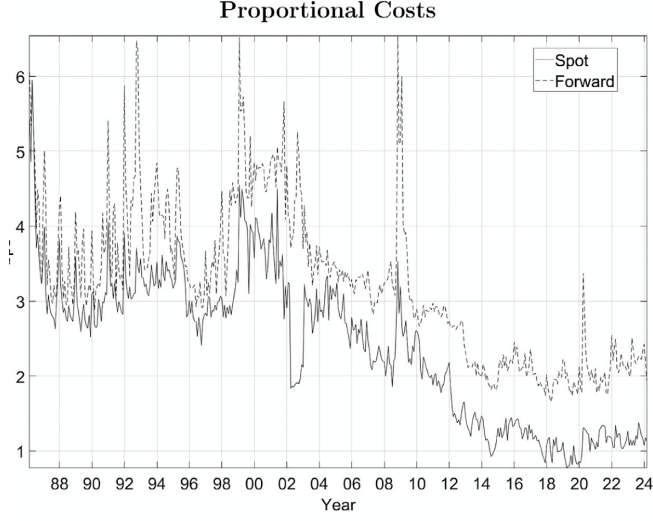


Figure 1: Proportional Costs. The solid line represents the cross-currency median of the average proportional costs (Section 2.3) for the spot market, $med(0.5C_{i,t}^{S+} + 0.5C_{i,t}^{P-})$.

(a) Figure 1: proportional costs.

	AUD	CAD	CHF	EUR	GBP	JPY
Hourly quotes	0.83	0.9	0.83	0.79	0.89	0.6
Constant volume	0.63	0.77	0.67	0.62	0.72	0.48

	MXN	NOK	NZD	SEK	SGD	ZAR
Hourly quotes	0.88	0.88	0.8	0.79	0.84	0.79
Constant volume	0.81	0.79	0.67	0.67	0.64	0.63

Notes: The table reports for the spot exchange rates of each currency $i \in I = \{\text{AUD, EUR, GBP, NZD, CAD, CHF, JPY, NOK, SEK, MXN, SGD, ZAR}\}$ the correlations between the daily realized Amihud measure of [Ranaldo and Santucci de Magistris \(2022\)](#) and two versions of our daily $A_{i,t}^S = \frac{RPV_{i,t}}{v_{i,t}^S}$ in Eq. (8) for the time period from September 2012 to September 2021. [Ranaldo and Santucci de Magistris \(2022\)](#) use one-minute exchange rate quotes to compute the daily return variation $RPV_{i,t}$ and daily volume data for $v_{i,t}^S$. In the row indicated by 'Hourly quotes' we use hourly exchange rate quotes to compute $RPV_{i,t}$ and daily volume data for $v_{i,t}^S$. In the row indicated by 'Constant Volume' we use hourly exchange rate quotes to compute $RPV_{i,t}$ and the time-series average of the daily volume (i.e., a constant) for $v_{i,t}^S$.

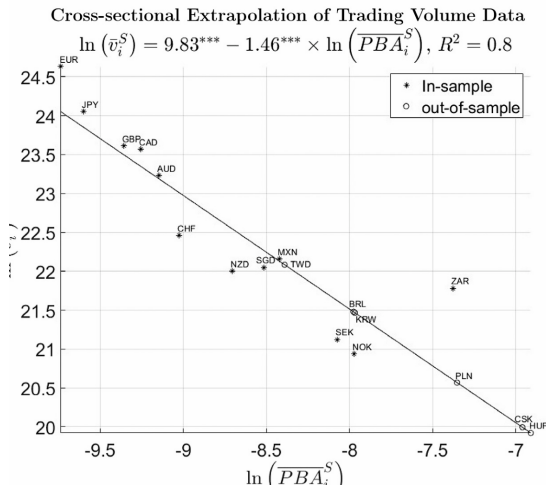
(b) Table 2: realized-Amihud correlations.

5.3 Figure 2 and Figure 3: Volume extrapolation and price impact

Read:

- Figure 2 plots average trading volume against average proportional bid-ask spreads.
- The regression slope is negative and the R^2 is about 0.8, supporting volume extrapolation.
- Figure 3 plots the cross-currency median of spot and forward price-impact costs.
- Price-impact costs spike during crisis periods and are visibly larger than proportional spreads.

Economic message: The paper's main cost channel is not the bid-ask spread but the effect of trade size on prices. The extrapolation makes it possible to study this channel over a long sample.



(a) Figure 2: cross-sectional volume extrapolation.

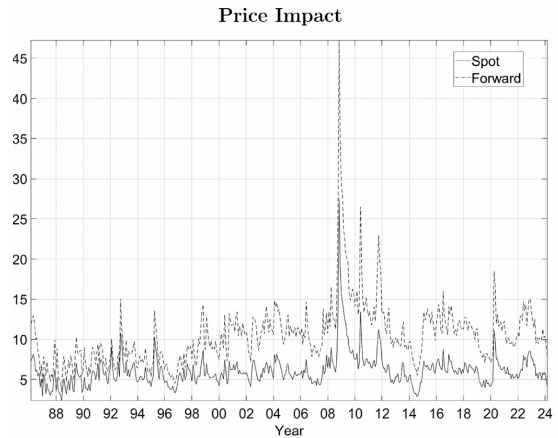


Figure 3: Price Impact. The solid line represents the cross-currency median of $A_{i,t}^S$ for spot (solid line) and $A_{i,t}^F$ for forwards (dashed line). $A_{i,t}^S$ and $A_{i,t}^F$ are the realized Amihud measures.

(b) Figure 3: price impact over time.

5.4 Table 3: Strategy performance before and after transaction costs

Read:

- With no price impact, *MV* and *MVTC* have Sharpe ratios around 1.0, while *HML* and *RB* are around 0.74–0.75.
- For AUM of USD 100 million, price impact reduces the *MV* Sharpe ratio to 0.36, but *MVTC* remains at 0.88.
- For AUM of USD 1 billion, *MV* becomes negative with a Sharpe ratio of -0.27, while *MVTC* remains positive at 0.75.
- Many simple strategies become weak or unprofitable at large scale because turnover and trade size create large price-impact costs.

Economic message: The table is the paper’s central result: transaction costs change the ranking of strategies. Accounting for costs in optimization is far more effective than merely deducting costs after trading.

Table 3
Performance of FX strategies.

	<i>MV</i>	<i>MVTC</i>	<i>HML</i>	<i>RB</i>	<i>DOL</i>	<i>DDOL</i>	<i>VAL</i>	<i>MOM</i>
Panel A: No price impact								
Before cost SR	1.12	1.08	0.79	0.81	0.09	0.32	0.51	0.36
Before cost mean (%)	8.92	8.40	6.55	5.50	0.70	2.45	3.50	3.15
Before cost Vol (%)	7.95	7.76	8.34	6.80	7.83	7.74	6.92	8.67
Mean costs (%)	1.06	0.58	0.34	0.39	0.02	0.19	0.11	0.30
Turnover	2.32	1.49	1.13	0.95	0.03	0.40	0.25	0.95
Notional	3.07	2.93	2.00	2.00	1.00	1.00	2.00	2.00
SR	0.99	1.01	0.74	0.75	0.09	0.29	0.49	0.33
ΔSR	-0.02	-	-0.07	-0.06	-0.92***	-0.32***	-0.68***	-
Mean (%)	7.85	7.81	6.20	5.11	0.68	2.25	3.39	2.85
Vol (%)	7.82	7.73	8.34	6.80	7.83	7.75	6.91	8.67
Skew	-0.83	-0.76	-0.77	-0.69	-0.41	-0.28	-0.20	-0.25
MDD (%)	-26.42	-28.02	-32.28	-31.43	-44.09	-18.30	-22.39	-54.73
Total return	44.56	44.23	23.73	16.32	2.37	4.87	8.25	6.13
Panel B: $AU/M_t = 100$ million								
Before cost SR	1.12	0.96	0.79	0.81	0.09	0.32	0.51	0.36
Before cost mean (%)	8.92	6.94	6.55	5.50	0.70	2.45	3.50	3.15
Before cost Vol (%)	7.95	7.25	8.34	6.80	7.83	7.74	6.92	8.67
Mean costs (%)	5.92	0.56	2.63	1.46	0.02	0.41	0.46	1.58
Turnover	2.32	0.68	1.13	0.95	0.03	0.40	0.25	0.95
Notional	3.07	2.99	2.00	2.00	1.00	1.00	2.00	2.00
Avg USD trade size (millions)	62.15	33.60	63.40	32.25	0.39	9.50	10.83	36.45
Avg relative trade size (%)	2.59	0.63	3.21	1.53	0.02	0.49	0.61	1.83
SR	0.36	0.88	0.46	0.59	0.09	0.26	0.44	0.18
ΔSR	-0.52***	-	-0.42**	-0.29*	-0.80***	-0.62***	-0.44**	-0.70***
Mean (%)	2.92	6.37	3.90	4.04	0.68	2.04	3.04	1.56
Vol (%)	8.14	7.22	8.39	6.83	7.83	7.77	6.92	8.69
Skew	-1.23	-1.03	-0.83	-0.71	-0.41	-0.29	-0.21	-0.24
MDD (%)	-35.91	-25.53	-33.24	-31.88	-44.10	-18.98	-22.55	-80.47
Total return	6.41	26.00	9.58	10.66	2.37	4.42	7.16	3.44
Panel C: $AU/M_t = 1$ billion								
Before cost SR	1.12	0.84	0.79	0.81	0.09	0.32	0.51	0.36
Before cost mean (%)	8.92	5.36	6.55	5.50	0.70	2.45	3.50	3.15
Before cost Vol (%)	7.95	6.41	8.34	6.80	7.83	7.74	6.92	8.67
Mean costs (%)	11.49	0.54	7.51	5.16	0.04	1.82	2.47	5.57
Turnover	2.32	0.29	1.13	0.95	0.03	0.40	0.25	0.95
Notional	3.07	2.16	2.00	2.00	1.00	1.00	2.00	2.00
Avg USD trade size (millions)	148.06	88.84	220.87	151.91	3.92	73.14	76.39	169.02
Avg relative trade size (%)	5.74	1.39	10.07	6.94	0.19	3.74	4.17	7.63
SR	-0.27	0.75	-0.12	0.04	0.08	0.08	0.14	-0.28
ΔSR	-1.02***	-	-0.86***	-0.70***	-0.66***	-0.67***	-0.61***	-1.02***
Mean (%)	-2.74	4.81	-1.01	0.51	0.60	0.63	1.02	-2.45
Vol (%)	10.03	6.44	8.82	7.09	7.83	7.99	7.20	8.86
Skew	-2.47	-0.95	-0.91	-0.83	-0.41	-0.42	-0.42	-0.24
MDD (%)	-157.21	-29.62	-115.01	-62.16	-44.14	-47.12	-38.37	-144.18
Total return	-0.14	14.49	0.71	1.94	2.34	2.20	2.91	0.00

Note: The table reports summary statistics of monthly excess returns of our strategies in the FX market. The strategies are described in Section 3. Panel A summarizes the performance when there are only proportional costs. Panel B and C report results when the initial AUM of a fund is USD 100 million respectively 1 billion and costs account for the price impact of trading. Before cost SR, Mean (%) and Vol (%) measure the annualized Sharpe ratio, and annualized average and volatility of the excess returns (reported in percentage points). Mean Costs (%) measures the average annualized trading costs as a percentage of the AUM. Turnover and Notional report the monthly turnover and the notional value as a fraction of the AUM. Avg USD Trade Size (millions) measures the average amount traded per month (average across currency-month observations with non-trivial trade) as a percentage of the average daily trading volume in the market respectively in absolute terms as millions of USD. SR is the annualized after cost Sharpe ratio. ΔSR is the difference in the after-cost Sharpe ratio of a strategy and that of *MVTC*. Standard errors of ΔSR are estimated using block bootstrapping with a block size of 5 observations to account for heteroskedasticity, cross- and auto-correlation (Cribari and Vail, 2010). Mean (%) and Vol (%) are the annualized average and volatility of the after cost excess returns (reported in percentage points). Skew is the monthly skewness of the after cost excess returns, MDD (%) is the Maximum Draw Down (measured in percentage points). Total return is the return (including the risk free rate) from February 1987 to January 2024. The sample includes 13 developed and 13 emerging currencies against the USD for the period from February 1987 to January 2024.

* Indicate a statistical significance at the 10% level.

** Indicate a statistical significance at the 5% level.

*** Indicate a statistical significance at the 1% level.

The Sharpe ratios of *DDOL* (0.32) and *MOM* (0.36) are lower than is the sample period. The former study uses data from 1983 to 2010,

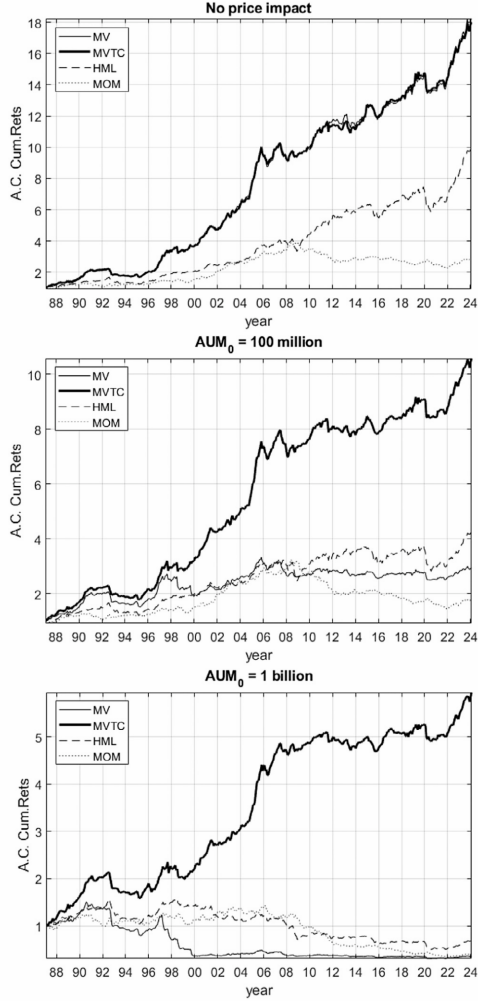
5.5 Figure 4 and Figure 5: Cumulative returns and strategy capacity

Read:

- Figure 4 shows cumulative after-cost excess returns for *MV*, *MVTC*, *HML*, and *MOM* under different AUM levels.
- *MVTC* remains robust as fund size increases, while other strategies flatten or deteriorate.
- Figure 5 plots after-cost Sharpe ratios as a function of initial AUM.
- Most strategies quickly fall below the SPY benchmark as AUM rises; *MVTC* remains above the benchmark over the plotted range.

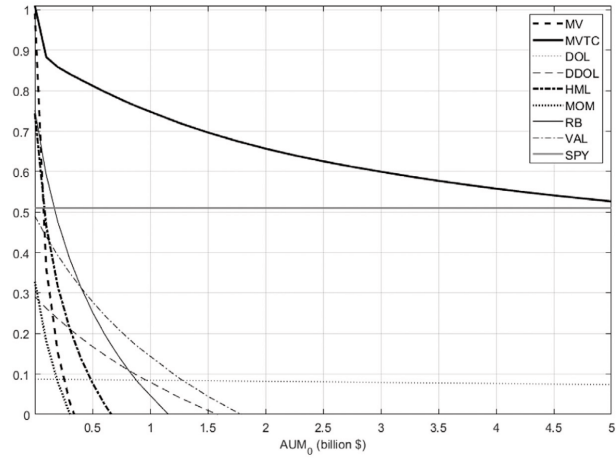
Economic message: Cost-aware portfolio design raises capacity. It is not enough to identify a profitable currency strategy; the strategy must also be scalable after accounting for trading impact.

Cumulative After-Cost Excess Returns



(a) Figure 4: cumulative after-cost returns.

Capacities of the Strategies: Effect of the Fund Size



after-cost Sharpe ratio as a function of the initial AUM of a fund for our FX trading strategies, and the US stock index (SPY). The sample includes 13 developed and 13 emerging currencies against the USD for the period 1986-2024.

(b) Figure 5: strategy capacity and AUM.

6 Short summary

- The paper studies transaction costs in FX asset allocation from 1986 to 2024.
- Bid-ask spread costs are modest, but price-impact costs from trading volume are economically large.
- The paper constructs a long realized-Amihud price-impact panel for 26 currencies.
- Volume data are extrapolated using the strong relation between bid-ask spreads and trading volume.
- Popular strategies such as carry, momentum, value, and mean-variance portfolios look attractive before price impact.
- Once price impact is included, many strategies become unprofitable at realistic institutional scale.
- *MVTC* internalizes transaction costs in portfolio choice and remains robust for much larger AUM.

- The main lesson is that FX strategy performance and capacity depend on trading costs, not only on before-cost expected returns.

7 Conclusion

7.1 Core conclusion

Filippou, Maurer, Pezzo, and Taylor show that transaction costs have a first-order effect on FX portfolio performance. Proportional bid–ask costs are not the main threat. The main threat is price impact, which grows with fund size and turnover.

7.2 Economic intuition

Currency markets are liquid, but large orders still move prices. A strategy that rebalances aggressively may look profitable before costs but lose its edge once trades are large enough to affect quotes. A cost-aware optimizer trades less, avoids expensive positions, and chooses a portfolio that is slightly less optimal before costs but much better after costs.

7.3 Takeaway

The paper shifts the evaluation of FX strategies from return prediction to implementability. For institutional investors, strategy capacity is determined by the interaction of expected returns, turnover, and price impact. *MVTC* is valuable because it solves this joint problem directly.